# Data Analysis Lab Solutions

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## Question 1: Dataset Summary & Overview

Load the `employee_salary_dataset.csv` dataset and perform basic exploratory data analysis including checking dimensions, data types, missing values, and statistical summary.

In [1]:
```
import pandas as pd

# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\employee_salary_dataset.csv")

# Number of rows and columns
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

# Data types
print("\nData Types:")
print(df.dtypes)

# Missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Statistical summary
print("\nStatistical Summary:")
print(df.describe(include='all'))
```

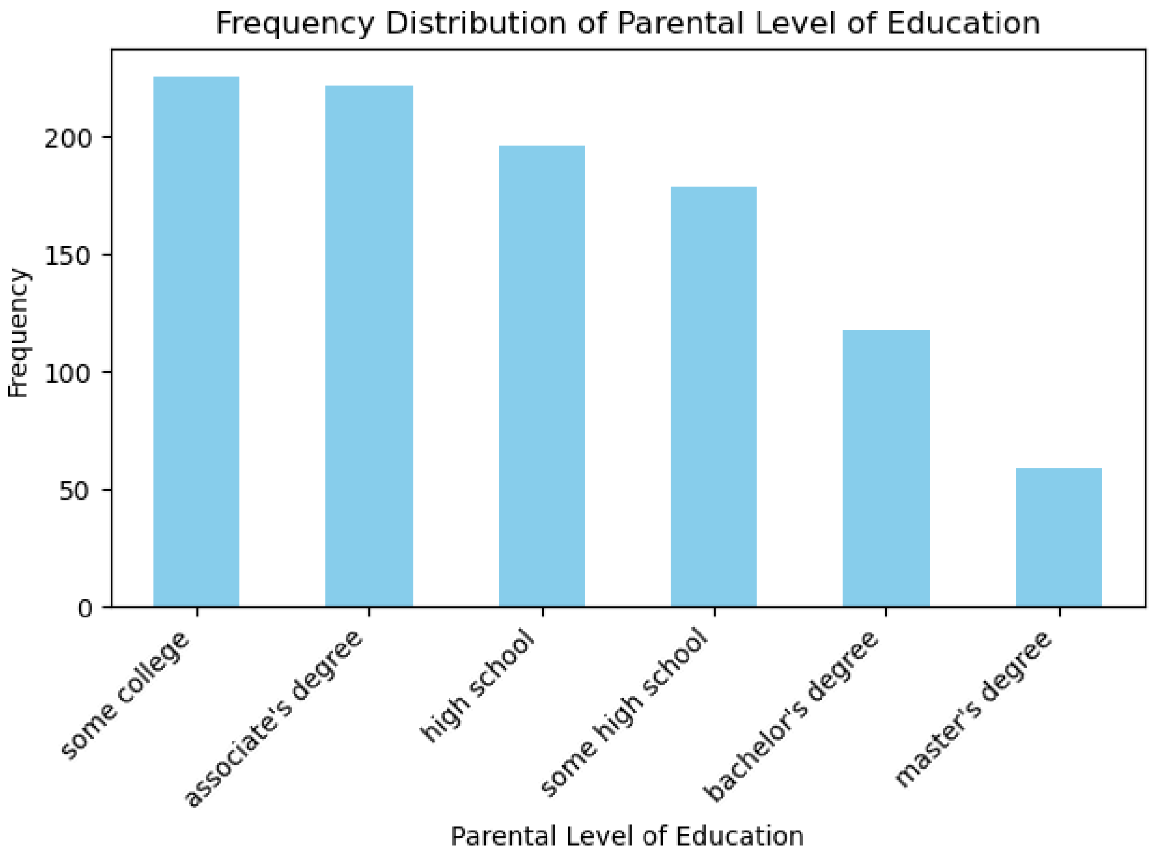
```
Number of Rows: 50
Number of Columns: 9

Data Types:
EmployeeID           int64
Name                object
Department          object
Experience_Years     int64
Education_Level     object
Age                  int64
Gender              object
City                object
Monthly_Salary       int64
dtype: object

Missing Values:
EmployeeID          0
Name                0
Department          0
Experience_Years    0
Education_Level     0
Age                 0
Gender              0
City                0
Monthly_Salary      0
dtype: int64

Statistical Summary:
        EmployeeID        Name Department  Experience_Years Education_Level  \
count     50.00000          50         50         50.000000              50
unique         NaN          50          5               NaN               4
top            NaN  Employee_1  Marketing               NaN          Master
freq           NaN           1         13               NaN              19
mean      25.50000         NaN        NaN          9.900000             NaN
std       14.57738         NaN        NaN          5.349995             NaN
min        1.00000         NaN        NaN          1.000000             NaN
25%       13.25000         NaN        NaN          5.250000             NaN
50%       25.50000         NaN        NaN         10.000000             NaN
75%       37.75000         NaN        NaN         14.750000             NaN
max       50.00000         NaN        NaN         19.000000             NaN

              Age  Gender   City  Monthly_Salary
count   50.000000      50     50         50.0000
unique        NaN       2      5             NaN
top           NaN  Female  Delhi             NaN
freq          NaN      27     15             NaN
mean    39.760000     NaN    NaN      82288.8000
std     11.539745     NaN    NaN      33521.4379
min     22.000000     NaN    NaN      28420.0000
25%     28.250000     NaN    NaN      59424.0000
50%     43.500000     NaN    NaN      73890.5000
75%     49.000000     NaN    NaN     107219.0000
max     57.000000     NaN    NaN     149123.0000
```

# Question 2: Categorical Frequency Distribution & Bar Chart

Load the `StudentsPerformance.csv` dataset, calculate the frequency distribution of parental level of education, and visualize it using a bar chart.

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\StudentsPerformance.csv")

# Frequency distribution
grade_freq = df["parental level of education"].value_counts()
print(grade_freq)

# Bar chart
grade_freq.plot(kind="bar", color="skyblue")
plt.title("Frequency Distribution of Parental Level of Education")
plt.xlabel("Parental Level of Education")
plt.ylabel("Frequency")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
parental level of education
some college          226
associate's degree    222
high school           196
some high school      179
bachelor's degree     118
master's degree        59
Name: count, dtype: int64
```

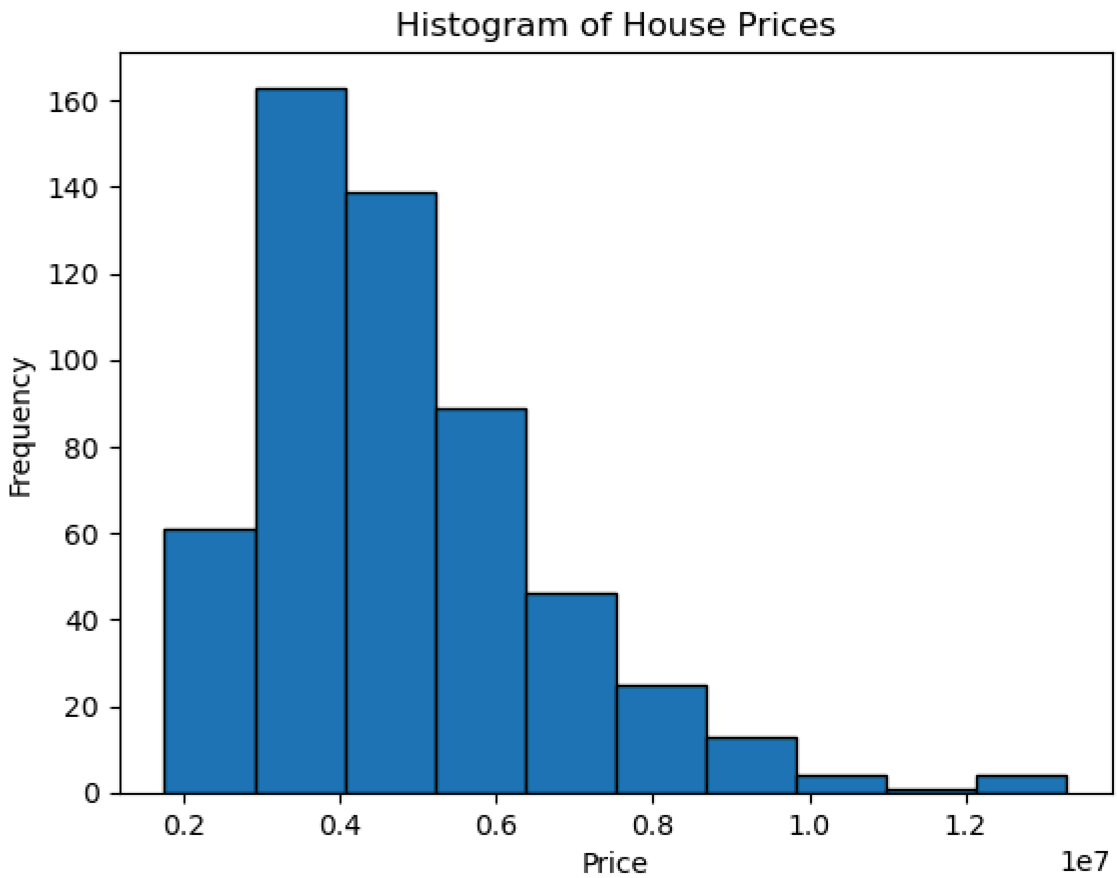



# Question 3: Histogram & Skewness Analysis

Load the `Housing.csv` dataset, plot a histogram for house prices, calculate skewness, and determine if the distribution is normal or skewed.

In [3]:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\Housing.csv")

# Histogram
plt.hist(df["price"], bins=10, edgecolor="black")
plt.title("Histogram of House Prices")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()

# Skewness
print("Skewness:", df["price"].skew())
```




```
Skewness: 1.2122388370279802
```

In [4]:

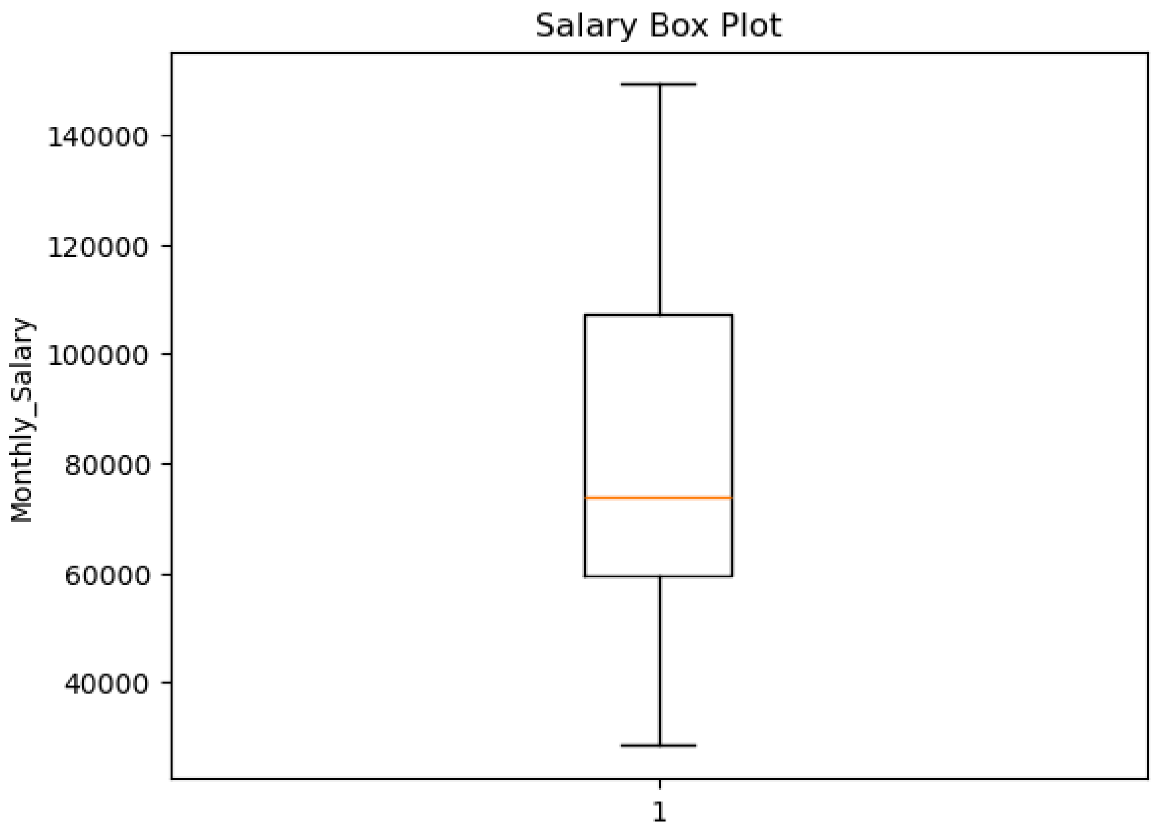
```
if abs(df["price"].skew()) < 0.5:
    print("Approximately Normally Distributed")
else:
    print("Skewed Distribution")
```

```
Skewed Distribution
```

# Question 4: Box Plot & Outlier Detection (Employee Salary)

Load the `employee_salary_dataset.csv` dataset, plot a box plot for salary, and detect outliers using the Interquartile Range (IQR) method.

In [5]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\employee_salary_dataset.csv")

# Box plot
plt.boxplot(df["Monthly_Salary"].dropna())
plt.title("Salary Box Plot")
plt.ylabel("Monthly_Salary")
plt.show()

# Outliers
Q1 = df["Monthly_Salary"].quantile(0.25)
Q3 = df["Monthly_Salary"].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Monthly_Salary"] < lower) | (df["Monthly_Salary"] > upper)]
print("Outliers:")
print(outliers)
```




```
Outliers:
Empty DataFrame
Columns: [EmployeeID, Name, Department, Experience_Years, Education_Level, Age, G
ender, City, Monthly_Salary]
Index: []
```

# Question 5: Outlier Identification in Student Marks

Load the `StudentsPerformance.csv` dataset and identify any outlier records based on student marks using the IQR method.

In [6]:
```
import pandas as pd

# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\StudentsPerformance.csv")

Q1 = df["math score"].quantile(0.25)
Q3 = df["math score"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["math score"] < lower) | (df["math score"] > upper)]
print("Outliers:")
print(outliers)
```

```
Outliers:
     gender race/ethnicity parental level of education         lunch  \
17   female        group B            some high school  free/reduced
59   female        group C            some high school  free/reduced
145  female        group C                some college  free/reduced
338  female        group B            some high school  free/reduced
466  female        group D          associate's degree  free/reduced
787  female        group B                some college      standard
842  female        group B                 high school  free/reduced
980  female        group B                 high school  free/reduced

    test preparation course  math score  reading score  writing score
17                     none          18             32             28
59                     none           0             17             10
145                    none          22             39             33
338                    none          24             38             27
466                    none          26             31             38
787                    none          19             38             32
842               completed          23             44             36
980                    none           8             24             23
```

# Question 6: Removing Outliers from Sales Data

Load the `Sample - Superstore.csv` dataset and remove outliers from the `Sales` column using the IQR technique to get a cleaned dataset.

In [7]:
```
import pandas as pd

# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\Sample - Superstore.csv", encoding="

Q1 = df["Sales"].quantile(0.25)
```

```
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned_df = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

print("Cleaned Dataset:")
print(cleaned_df.head())
```

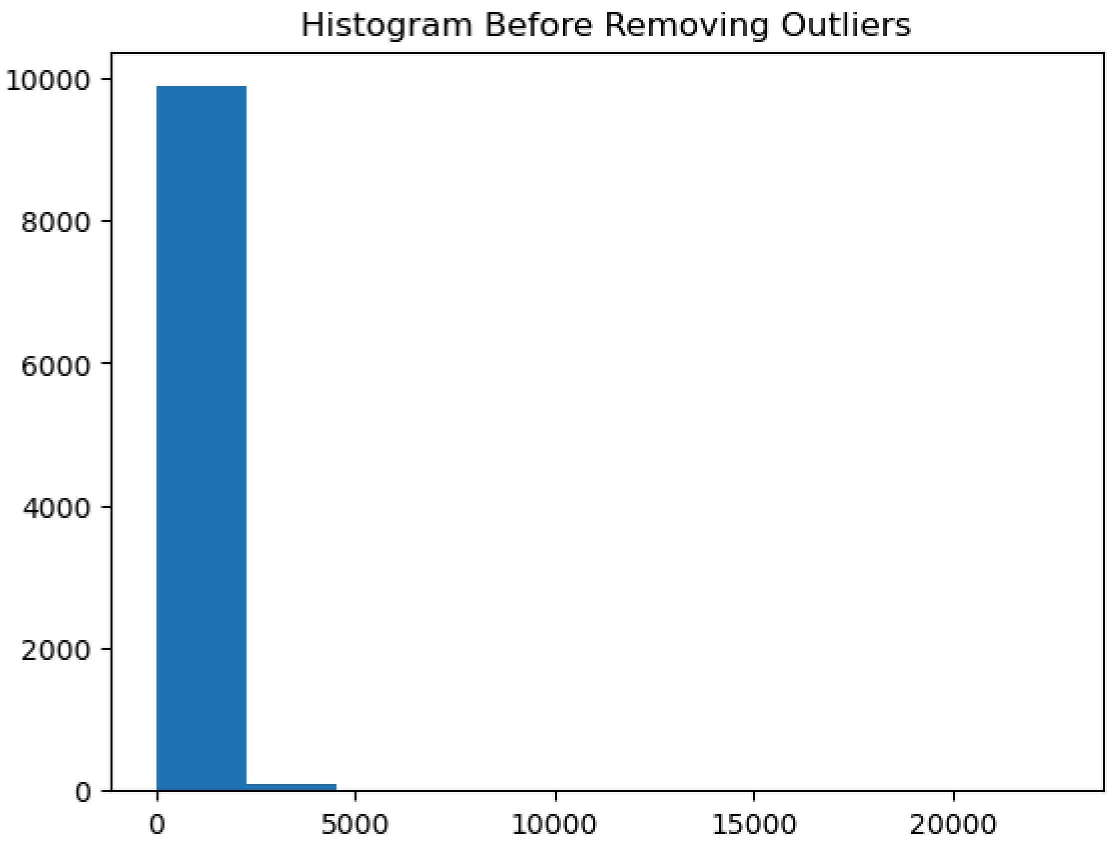
```
Cleaned Dataset:
   Row ID        Order ID  Order Date   Ship Date       Ship Mode Customer ID  \
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class    DV-13045
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335
5       6  CA-2014-115812    6/9/2014   6/14/2014  Standard Class    BH-11710
6       7  CA-2014-115812    6/9/2014   6/14/2014  Standard Class    BH-11710

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
5  Brosina Hoffman   Consumer  United States      Los Angeles  ...
6  Brosina Hoffman   Consumer  United States      Los Angeles  ...

  Postal Code  Region       Product ID         Category Sub-Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases
2       90036    West  OFF-LA-10000240  Office Supplies       Labels
4       33311   South  OFF-ST-10000760  Office Supplies      Storage
5       90032    West  FUR-FU-10001487        Furniture  Furnishings
6       90032    West  OFF-AR-10002833  Office Supplies          Art

                                        Product Name    Sales  Quantity  \
0                  Bush Somerset Collection Bookcase  261.960         2
2  Self-Adhesive Address Labels for Typewriters b...   14.620         2
4                      Eldon Fold 'N Roll Cart System   22.368         2
5  Eldon Expressions Wood and Plastic Desk Access...   48.860         7
6                                         Newell 322    7.280         4

   Discount   Profit
0       0.0  41.9136
2       0.0   6.8714
4       0.2   2.5164
5       0.0  14.1694
6       0.0   1.9656

[5 rows x 21 columns]
```

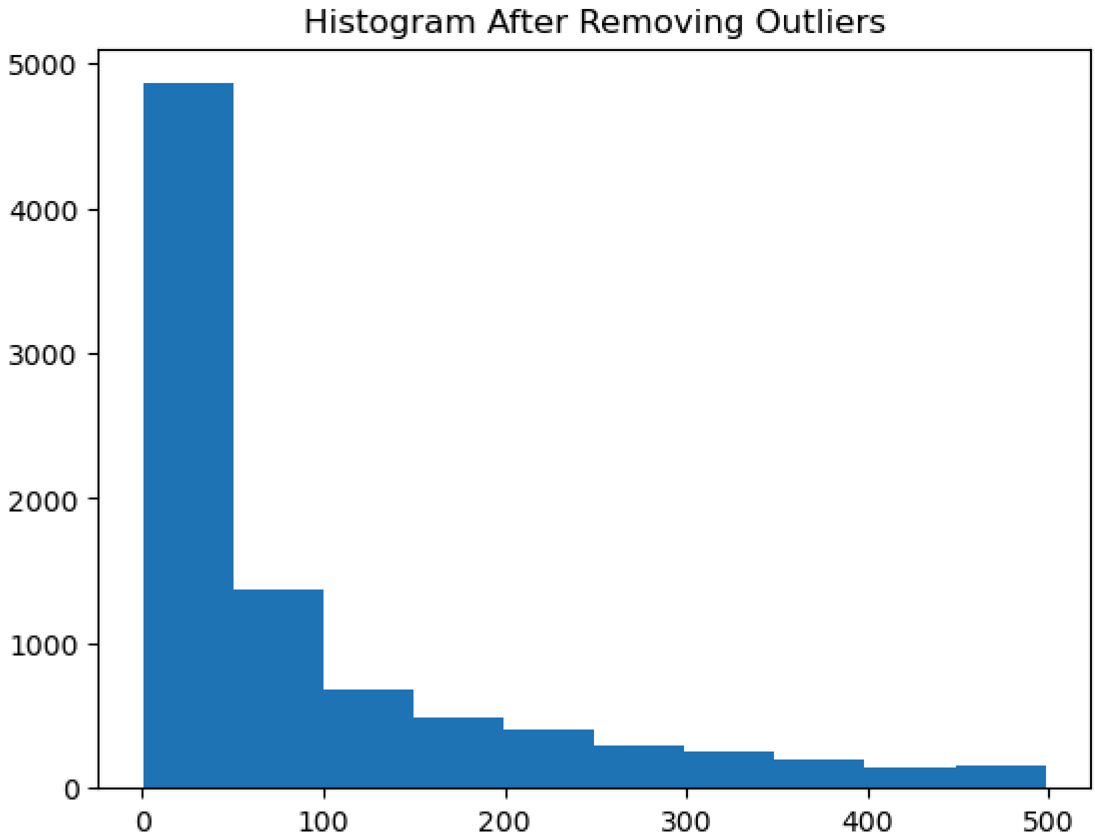
# Question 7: Visualization Comparison Before & After Outlier Removal

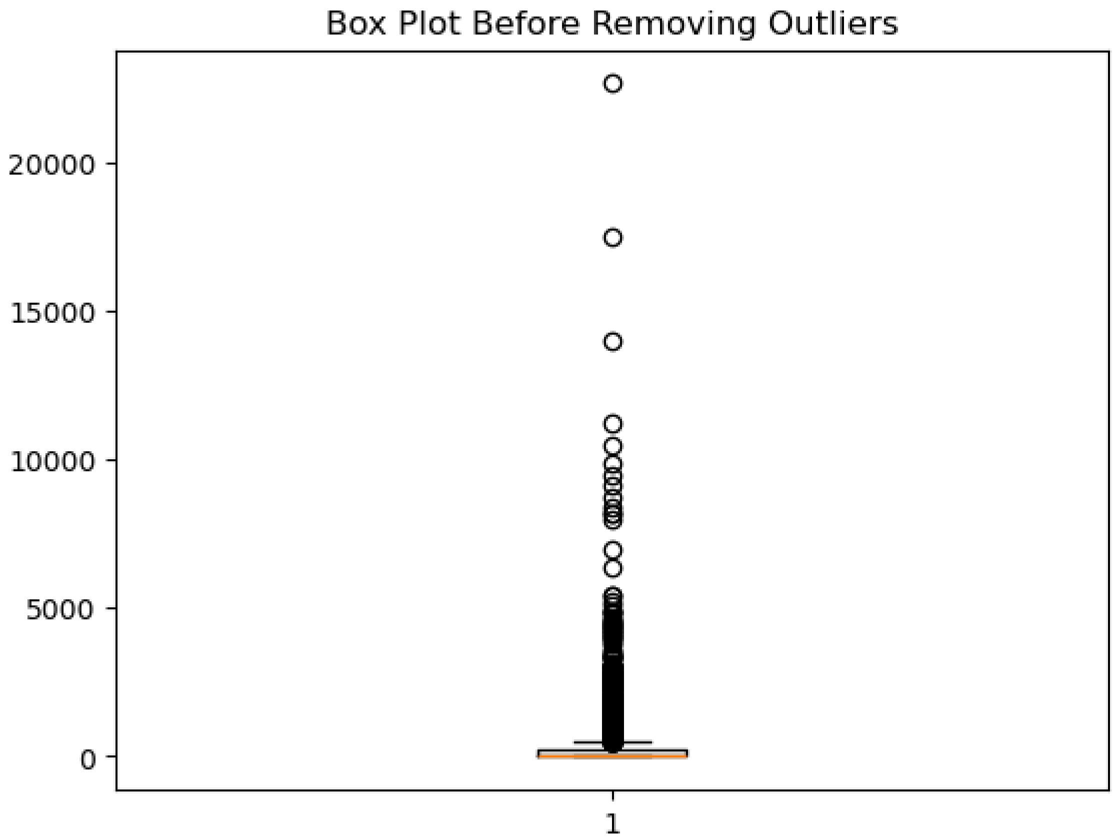
Load `Sample - Superstore.csv`, filter out the outliers using IQR, and plot histograms and box plots before and after outlier removal for visual comparison.

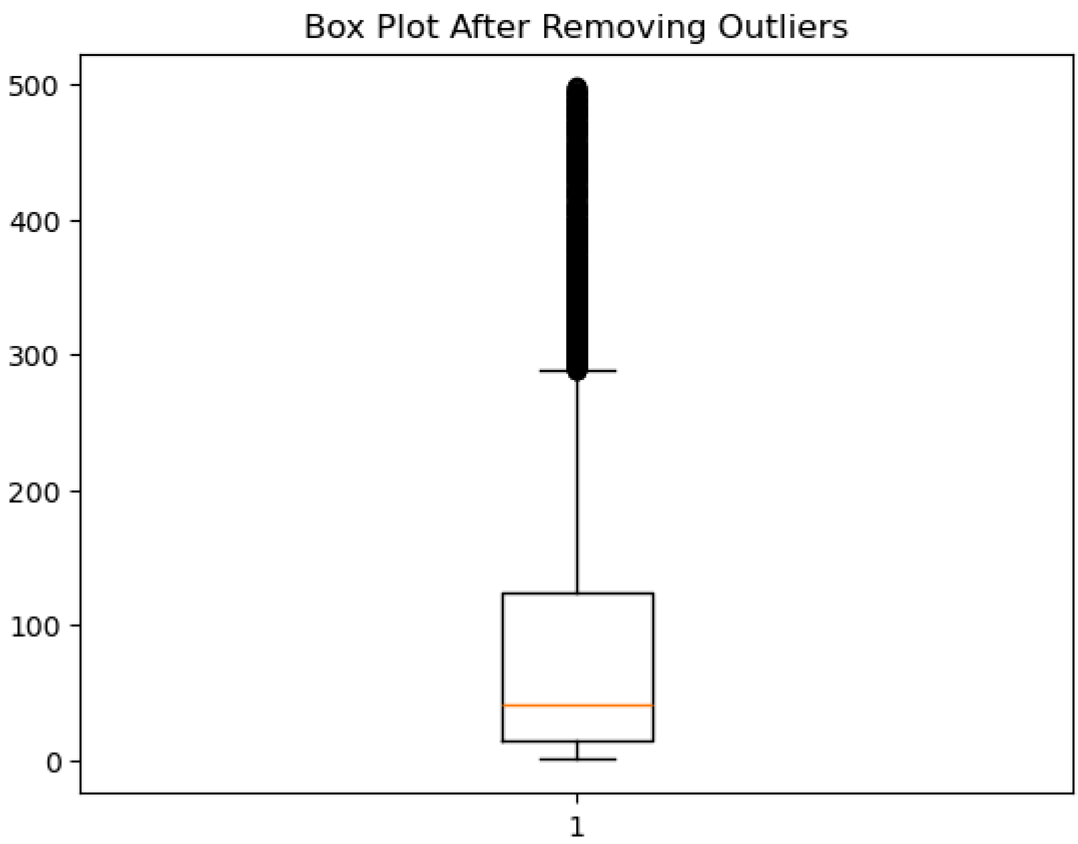
In [8]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Load dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\Sample - Superstore.csv", encoding="

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

# Histogram Before
plt.hist(df["Sales"], bins=10)
plt.title("Histogram Before Removing Outliers")
plt.show()

# Histogram After
plt.hist(cleaned["Sales"], bins=10)
plt.title("Histogram After Removing Outliers")
plt.show()

# Box Plot Before
plt.boxplot(df["Sales"])
plt.title("Box Plot Before Removing Outliers")
plt.show()

# Box Plot After
plt.boxplot(cleaned["Sales"])
plt.title("Box Plot After Removing Outliers")
plt.show()
```




Histogram After Removing Outliers
5000
4000
3000
2000
1000
0
0
100
200
300
400
500


Box Plot Before Removing Outliers
20000
15000
10000
5000
0
1

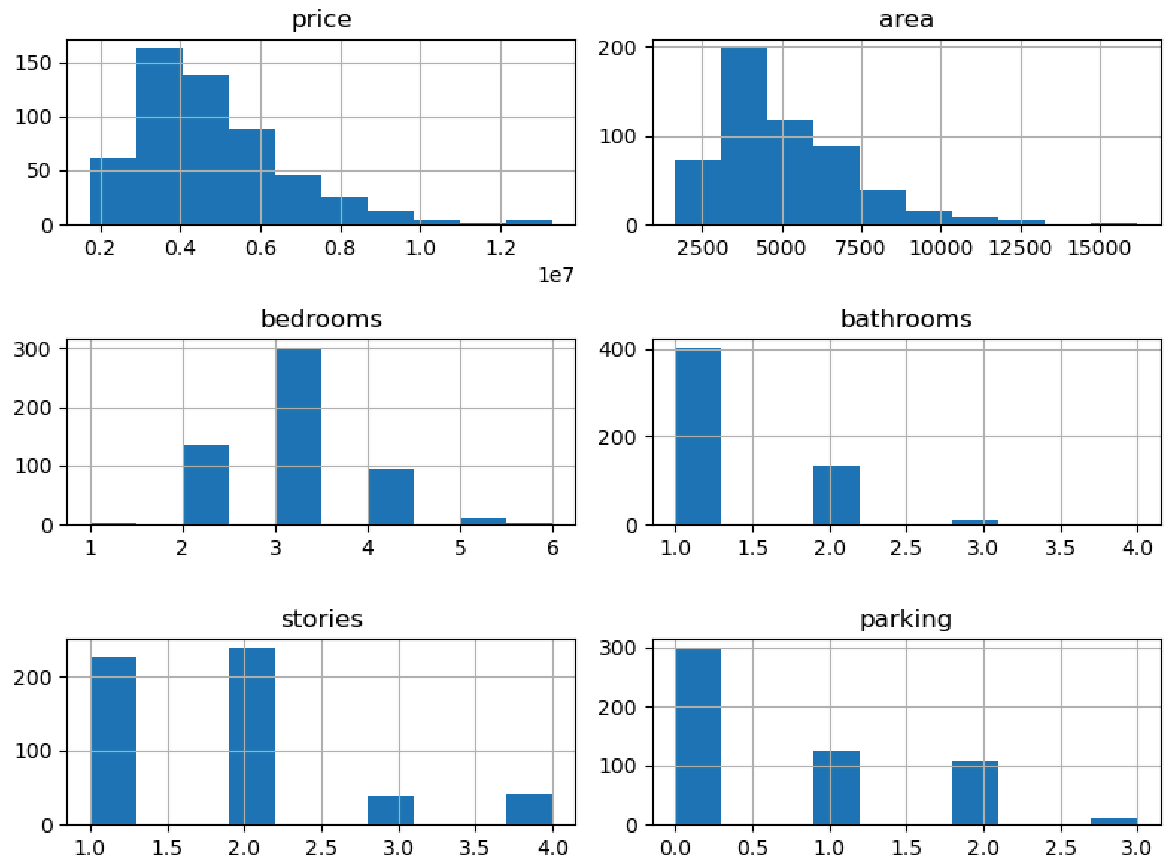



## Question 8: Full Exploratory Data Analysis & Outlier Cleaning

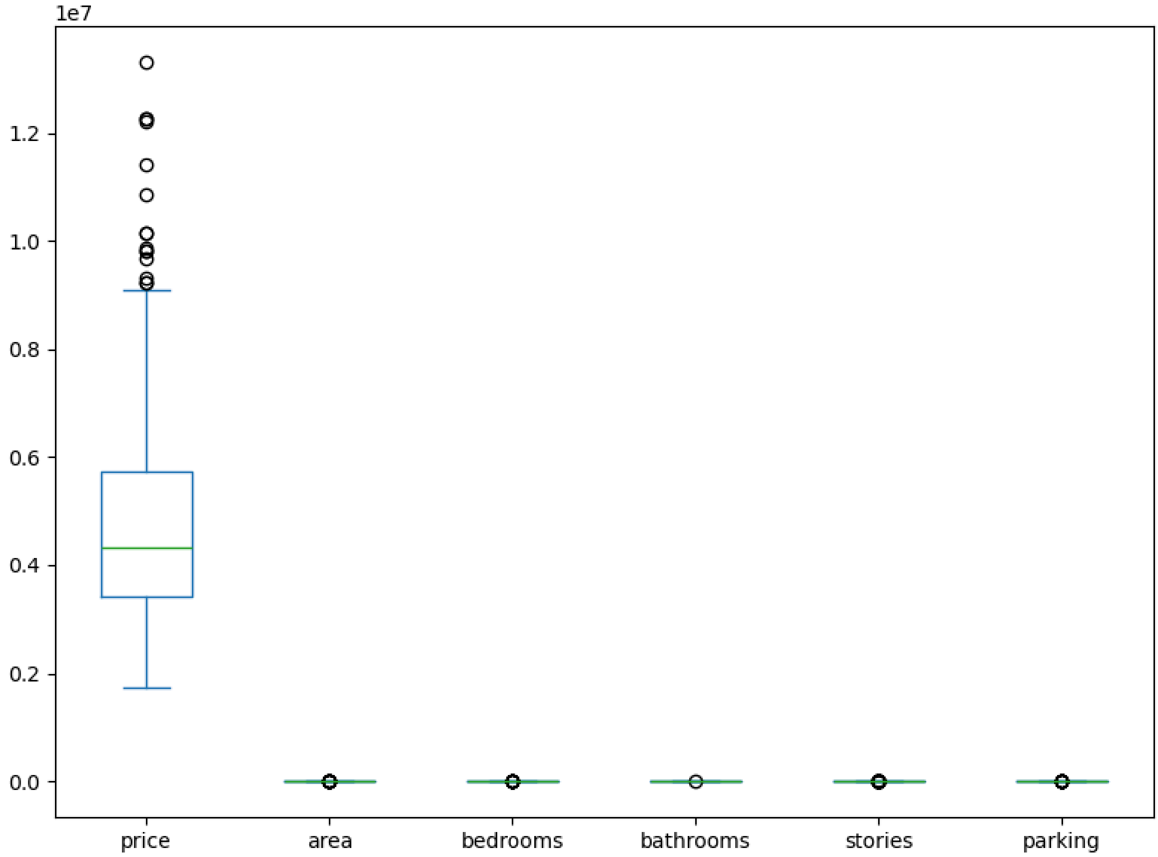
Load the dataset, check missing values and descriptive statistics, plot histograms and box plots, remove numerical outliers across all features, and save the cleaned dataset.

In [9]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Load Housing dataset
df = pd.read_csv(r"C:\Users\btaru\Downloads\Housing.csv")

# Dataset Information
print("Dataset Info:")
print(df.info())

# Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# Descriptive Statistics
print("\nDescriptive Statistics:")
print(df.describe())

# Histograms
df.select_dtypes(include='number').hist(figsize=(8,6))
plt.tight_layout()
plt.show()
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype 
---  ------            --------------  ----- 
 0   price             545 non-null    int64 
 1   area              545 non-null    int64 
 2   bedrooms          545 non-null    int64 
 3   bathrooms         545 non-null    int64 
 4   stories           545 non-null    int64 
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64 
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
None

Missing Values:
price               0
area                0
bedrooms            0
bathrooms           0
stories             0
mainroad            0
guestroom           0
basement            0
hotwaterheating     0
airconditioning     0
parking             0
prefarea            0
furnishingstatus    0
dtype: int64

Descriptive Statistics:
              price          area    bedrooms   bathrooms     stories  \
count  5.450000e+02    545.000000  545.000000  545.000000  545.000000   
mean   4.766729e+06   5150.541284    2.965138    1.286239    1.805505   
std    1.870440e+06   2170.141023    0.738064    0.502470    0.867492   
min    1.750000e+06   1650.000000    1.000000    1.000000    1.000000   
25%    3.430000e+06   3600.000000    2.000000    1.000000    1.000000   
50%    4.340000e+06   4600.000000    3.000000    1.000000    2.000000   
75%    5.740000e+06   6360.000000    3.000000    2.000000    2.000000   
max    1.330000e+07  16200.000000    6.000000    4.000000    4.000000   

          parking  
count  545.000000  
mean     0.693578  
std      0.861586  
min      0.000000  
25%      0.000000  
50%      0.000000  
75%      1.000000  
max      3.000000  
```




In [10]:
```
# Box Plots
df.select_dtypes(include='number').plot(kind="box", figsize=(8,6))
plt.tight_layout()
plt.show()

# Detect and Remove Outliers
numeric_cols = df.select_dtypes(include='number').columns

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df = df[(df[col] >= lower) & (df[col] <= upper)]

print("\nCleaned Dataset:")
print(df.head())

# Save Cleaned Dataset
df.to_csv("housing_cleaned.csv", index=False)
print("Cleaned dataset saved as housing_cleaned.csv")
```




```
Cleaned Dataset:
      price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
15  9100000  6000         4          1        2      yes        no      yes
20  8750000  4320         3          1        2      yes        no      yes
22  8645000  8050         3          1        1      yes       yes      yes
27  8400000  8875         3          1        1      yes        no       no
40  7875000  6550         3          1        2      yes        no      yes

   hotwaterheating airconditioning  parking prefarea furnishingstatus
15              no              no        2       no   semi-furnished
20             yes              no        2       no   semi-furnished
22              no             yes        1       no        furnished
27              no              no        1       no   semi-furnished
40              no             yes        0      yes        furnished
Cleaned dataset saved as housing_cleaned.csv
```